Anthony Ibarra | Electrical & Computer Engineer

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Summary

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective career. Showcasing a strong focus in Embedded Systems, Control theory, Robotics, including pattern recognition involving AI CNN machine learning. And power electronics making me a flexible Embedded Software Engineer with a strong foundation. I possess various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness to depend on one another.

EDUCATION

University of Illinois Chicago (UIC) — Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering

Relevant Coursework:

Audio & Acoustic Signal Processing, Intro to Filter Synthesis, Adaptive Digital Filters, Electromechanical Energy Conversion, Linear Systems Theory & Design, Electromagnetic Compatibility, Advanced Computer Communication, Convex Optimization, Mechatronics Embedded Design, Intro Neural Networks

University of Illinois Chicago (UIC) — Aug 2017 - May 2022

Bachelor of Science in Computer Engineering

Relevant Coursework:

Artificial Intelligence I, Principles of Modern Control & Principles of Auto Control, Pattern Recognition I, Computer Comm Networks I, Robotics: Algorithm/Control, Embedded Systems

SKILLS

Computer programming: Python, C/C++, verilog, VHDL, object oriented programming, Assembly, Ubuntu, AI (artificial intelligence) & ML(machine learning), ROS2, PLC, HMI SCADA

Software Knowledge: VS Code, GitHub, MATLAB, Solidworks, Mathematica, Altium, Ltspice, HFSS Ansys simulator, Quartus, code-composer, Arduino IDE, VNC viewer(raspberry Pi), (Microsoft Word, PowerPoint, Excel, etc.), Rockwell Automation RSLogix Rslinx, STM32 CubeMX, CCW (connected component workbench)

Speach: B1 Proficient in spanish

Projects

Research Project, Acoustic & Audio Signal Processing — Jan 2025 - May 2025

Implement a solution for motorcycle riders with some degree of hearing impairment. The starting goal is to remove the dampening from the helmet, providing clear sound while wearing the helmet without adding any unneeded noise or unwanted sounds. In turn this can potentially amplify user awareness while riding or in largely populated areas. Report being in the scope and style of a Signal Processing Society conference paper.

- Develop a new method or solution with concepts to be applied to an existing problem. Perform real world testing and validation of project and research implementation
- Test and use skills and algorithms from acoustic & audio signal processing to achieve the set goal

Optimization on Three Coil Long Range (WPT), Electromagnetic Compatibility — Jan 2024 - May 2024

- Electromagnetic Compatibility testing using a spectrum analyzer, Oscilloscopes for EMC and EMI testing for PCB and signal integrity using RF SMA coaxial cables, SMA terminators
- Create a three coil configuration for long range wireless power transfer for simulation, confirming data from research papers through simulations
- Mathematically simulated plot from Matlab regarding the Three coils as shown within the research paper
- Simulation and plot from Ansys HFSS 3D EM simulation software, simulating the three coil behavior in different configurations
- Make an IEEE formatted report using Latex, in a professionally typeset manner demonstrating discoveries and results

Self Driving Car, Mechatronics Embedded Design
Jan 2023 - May 2023

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary
- Develop and tune controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

Automated Watering System, Senior Design
August 2021 - May 2022

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device